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RESEARCH ARTICLE

Integration of Emotional Factors With GAN Algorithm in Stock Price Prediction Method Research

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ABSTRACT Stock prices are characterized by non-stationarity and volatility, and investors are easily influenced by their own emotions, and their investment decision-making behavior is characterized by irrationality, so stock prices are difficult to predict. This paper proposes a stock price prediction method based on a generative adversarial network combining sentiment factors with financial data. The method is based on the GAN algorithm, which better matches the logic behind the operation of the stock market, and introduces the text branch to realize the sentiment analysis of the text of the research report, which provides the public opinion influence factor for the algorithmic model, and proposes the TK-GAN model. The model combines financial data with sentiment factors to reduce the error of model training, and introduces Soft Attention to improve the learning ability of stock-related data features. In addition, the TK-GAN model utilizes BERT for financial domain fine-tuning in the text branch to increase the model's suitability for specific financial domains. Based on the TK-GAN model, AdamK, a learning rate adaptive optimization algorithm, which is more suitable for this research, is proposed. The experimental results of the model on the Kweichow Moutai stock dataset show that the MAE and MSE values are as low as 0.01949 and 0.00091 respectively.

INDEX TERMS Stock prediction, sentiment factors, GAN, soft attention, optimizer.

I. INTRODUCTION

With the rapid development of the global economy, stocks, as a kind of high-risk and high-yield investment subject, are deeply concerned by the majority of investors. Traditional stock prediction methods are based on econometrics, and in the 1980s, with the emergence of Boltzmann machine and backpropagation (BP) algorithm [1], BP neural network was used for stock market modeling and decision making, which proved the existence of practical value of BP neural network for stock prediction [2], [3], [4].

Subsequently, Hochreiter and Schmidhuber [5] proposed the LSTM algorithm, which demonstrated unique advantages in time series problems. Sun [6] used the LSTM network in stock prediction research, and verified through experimental comparisons that the LSTM network was more accurate than the BP network model in stock prediction. Deng and Wang [7] utilized the LSTM network for prediction, and the results showed that the LSTM model can maintain high prediction accuracy in U.S. stock prediction. Wang et al. [8] combined market data with technical data, downscaled the data by principal component analysis, and then predicted the stock data by LSTM model, which greatly improved the prediction accuracy and reduced the runtime compared with the traditional LSTM. Deng et al. [9] used the LSTM model with a data of a certain sequence length as a test set, through this test set to predict the closing price of the next point; and then add the closing price of this point to the test set, and the cycle

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is repeated, proving that multi-day data in comparison with single-day data in the prediction of the accuracy rate obtained is higher.

Due to its excellent performance, The attention mechanism technology used in the image field is gradually being used in financial time series such as stocks. Zheng and Xu [10]came up with a model to predict stock price trend using one kind of improved self-attention mechanism, which mixes and encodes daily data time-line data respectively, and learns the effect of capital flow changes on the stock trend, which substantially improves the stock prediction. Chen and Liu [11] proposed the LSTM-CNN model based on the attention mechanism, and experimentally proved that this model has excellent generalization in stock trend prediction. Gu et al. [12] proposed the GRU model based on the attention mechanism for prediction, and approved the model performance was significantly improved by the attention mechanism to capture the important features in the time dimension. Zhang et al. [13] combined the empirical modal decomposition on top of the GRU model with an attention mechanism to further improve the results. Li et al. [14] brought up a Transformer-based Attention Model which consists of a feature extractor and a cascade processor, and uses the research text and stock price information of five days from the trading day as the training data, which reveals the correlation between the research text and the financial data that are strongly correlated.

Another idea of stock price prediction is to simulate the logic behind the operation of the stock market through the game idea in Generative Adversarial Networks [15] (GANs), which in turn enables stock price prediction. Zhang et al. [16] proposed a GAN architecture that uses MLP as a discriminator and LSTM as a generator for predicting the closing price of a stock and tried to predict the daily closing price. Zhou et al. [17] proposed a stock prediction model GAN-FD that uses thirteen key factors of stock as input data, CNN as an adversarial training, and LSTM as a generator to predict the stock price direction. a stock prediction model, GAN-FD, which uses thirteen key factors of stock as input data, CNN as discriminator for adversarial training, and LSTM as generator to predict stock price direction. Wang et al. [18] combined the empirical modal stock data decomposition method to solve the problem of instability of the traditional GAN prediction model and achieved better prediction in the existing stock dataset. effect. Faraz and Khaloozadeh [19] used least squares loss function on the basis of GAN model and preprocessed the data with wavelet transform to reduce the effect of stock market noise on the model performance and the model (LSGAN) outperformed GAN on S&P500 index. Polamuri et al. [20] proposed a hybrid prediction algorithm based on generative adversarial network (GAN- HPA). The algorithm optimizes the GAN parameters through reinforcement learning and Bayesian algorithm, and uses LSTM and CNN to implement the generator and discriminator respectively with good results.

Based on the above studies, we can find few studies that combine investors' emotional factors and financial data to realize stock prediction. Experiments have shown that methods that fuse text information and stock prices within a small time window outperform methods that use text or stock price generation alone [21], [22]. However, few studies have used this approach to predict stock prices; therefore, it is valuable to propose a research method that combines textual information with financial data. Therefore, this paper proposes a stock price prediction method based on generative adversarial networks and combining sentiment factors with financial data. The sentiment analysis of the research text provides the opinion influence factor for the algorithmic model, and the GAN algorithm, which better matches the logic behind the operation of the stock market, is introduced.

II. STOCK PREDICTION MODEL (TK-GAN)

A. TK-GAN MODEL FRAMEWORK

In the field of deep learning, one of the popular research directions is Generative Adversarial Networks (GAN) [23], [24], [25], [26], [27], [28], [29]. GAN refers to a class of neural networks designed based on game theory, rather than specifically indicating a particular neural network structure. It consists of a Generator (G) and a Discriminator (D), where the discriminator is responsible for distinguishing between real and fake data, and the generator generates simulated data closely resembling real samples. During training, the generator and discriminator engage in a game, and when the discriminator cannot distinguish between the two types of samples, the generator can better capture the distribution of real sample data.

The TK-GAN algorithm proposed in this paper is based on the GAN network structure, using GAN to simulate stock trends. Due to the wealth of information in stock text data, the TK-GAN algorithm introduces a Text Pathway in the generator, incorporating text-related parameters. This integrates the text analysis model of research reports with the GAN model, sharing training parameters and training errors, thereby achieving better stock prediction results. The overall model structure of TK-GAN is illustrated in FIGURE 1.

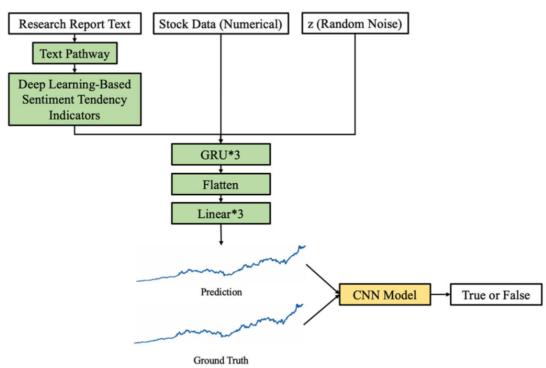

FIGURE 1. TK-GAN model structure diagram.

1) GENERATOR MODEL

In the GAN model, the generator G takes random noise z as input and, through continuous training, generates synthetic data G(z) with the same dimensions and a similar distribution as real data. The generator is essentially a neural network that, by learning the parameters of the data distribution, regenerates new data. In this process, the generator neural network needs to make distribution assumptions about the explicit or implicit variables of the data. It then inputs real data into the model for training optimization, ultimately obtaining a learned approximate distribution used for generating new data.

The generator network of TK-GAN adopts two types of inputs: preprocessed research report text and numerical stock data. The preprocessed research report text is input into the Text Pathway to obtain deep learning sentiment tendency factors. The combination of sentiment tendency factors, numerical stock data, and random noise z constitutes the complete input of the generator network. The generator network consists of BiLSTM and GRU, both of which have good performance in handling sequential data. Additionally, a soft attention mechanism is added to allow the generator to focus more on features that are more important for prediction. Finally, a fully connected neural network is used as the output layer to generate target data. Through this structural design, the paper aims to train a model with powerful generative capabilities, thereby improving the accuracy of predicting stock price trends.

2) DISCRIMINATOR MODEL

The discriminator D is a binary neural network classifier in the GAN model, used to determine the authenticity of input data rather than distinguish the original categories of the input data. For input data $\hat{x}$, the discriminator D outputs a probability value $D(\hat{x})$ between 0 and 1, where $\hat{x}$ may be a real data point from the real dataset or a synthetic data point G(z) generated by the generator G. The closer the probability value $D(\hat{x})$ is to 1, the more likely the data is real. Conversely, the smaller the probability value, the less likely $\hat{x}$ is real data.

The discriminator of TK-GAN is mainly composed of 1D CNN and a fully connected neural network, as shown in TABLE 1. The role of CNN is to extract features from input time series data. By stacking multiple convolutional layers, the discriminator can extract key information from the input data as the basis for discrimination. Subsequently, through a fully connected neural network, this high-dimensional feature information is further calculated, and a score is output as an evaluation criterion for the authenticity of the current input data by the discriminator.

3) OBJECTIVE FUNCTION

The loss function of the TK-GAN model is represented as shown in Equation 1, where G denotes the generator in the GAN network, and D represents the discriminator.

$$min_G max_D F(G, D) = E_{y\sim P_{data}}\left[log\left(D\left(y\right)\right)\right] + E_{y\sim P_G}\left[log\left(1\text{-D}\left(y\right)\right)\right] \quad (1)$$

In the equation: $y\sim P_{data}$ represents samples from the real data distribution, $E_{y\sim P_{data}}\left[log\left(D\left(y\right)\right)\right]$ signifies that when y is sampled from P_{data}, the output $D\left(y\right)$ through the discriminator should be maximized. $y\sim P_G$ indicates that y is from generated samples P_G, and $E_{y\sim P_G}\left[log\left(1\text{-D}\left(y\right)\right)\right]$ implies that when y is from P_{data}, the output $D\left(y\right)$ through the discriminator should be minimized.

TABLE 1. Discriminator structure of TK-GAN.

Network layer	Network layer parameter
Conv1d（LeakyReLU）	in_channels=1; out_channels=128; Kernel=3; stride=1
Conv1d（LeakyReLU）	in_channels=128; out_channels=256; Kernel=3; stride=1
Conv1d（LeakyReLU）	in_channels=256; out_channels=128; Kernel=3; stride=1
Flatten	/
Linear（LeakyReLU）	in_channels=128*4; out_channels=512
Linear（LeakyReLU）	in_channels=512; out_channels=512
Linear（Sigmoid）	in_channels=512; out_channels=1

B. TEXT PATHWAY

Investor sentiment is often mentioned in economic activities, reflecting the investment intentions and expectations of market participants, which can influence subsequent investment and decision-making [30]. In this study, we utilize the text of stock-related research reports in the TK-GAN generator to construct the Text Pathway. This pathway is designed to extract public sentiment factors, introducing the impact of public sentiment on the stock market to assist in stock trend prediction. The specific model structure of the Text Pathway is illustrated in TABLE 2.

TABLE 2. Structure of the text pathway.

Network layer	Network layer parameter
Embedding	embedding dim=128
BiLSTM	hidden_size=32, batch_first=True
Soft Attention	/
Linear	In features=64, out_features=48
Sigmoid	/

The preprocessed research report text undergoes fine-tuning with BERT to obtain BERT Embeddings. Subsequently, it passes through a BiLSTM layer, a linear layer, and a sigmoid activation function layer to output the predicted sentiment tendency factor. The Text Pathway utilizes a synchronized training approach by adding a text pathway to the GAN network. This allows for simultaneous updating of the sentiment tendency feature extraction network during stock prediction. Consequently, the two independent prediction processes are integrated into a synchronized and unified network structure, reducing training errors. The optimization of model training and error integration is visualized in the model structure, as shown in FIGURE 2.

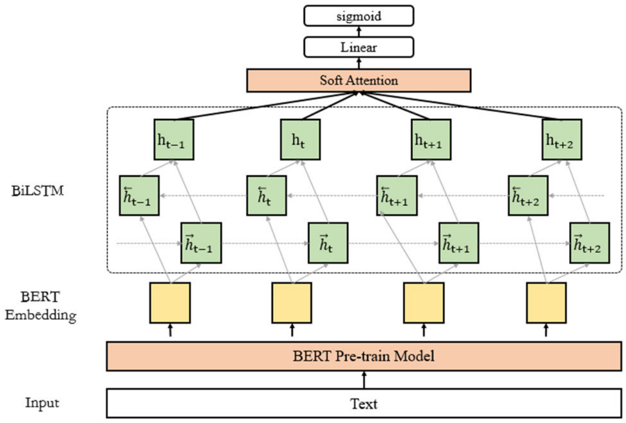

FIGURE 2. Text pathway model structure.

C. SOFT ATTENTION MECHANISM

Different parts of the human retina have varying degrees of information processing capabilities, and visual information processing resources are limited. To efficiently use these visual information processing resources, humans need to selectively focus on specific content in a visual area while ignoring other less important information. This mechanism of selectively attending to information is commonly referred to as the Attention mechanism [31], [32], [33], [34], [35]. In computer vision, the fundamental idea behind the attention mechanism is to enable neural networks to focus on important information and ignore irrelevant information. The concept of Soft Attention was first proposed by Xu et al. in 2015 [36] and has since achieved significant results.

By using the Attention mechanism, it is possible to assign higher weights to key features in the fused features, thereby improving the model's recognition performance. Therefore, in the Text Pathway, Soft Attention is applied to enhance the weights of key features, as illustrated in FIGURE 3.

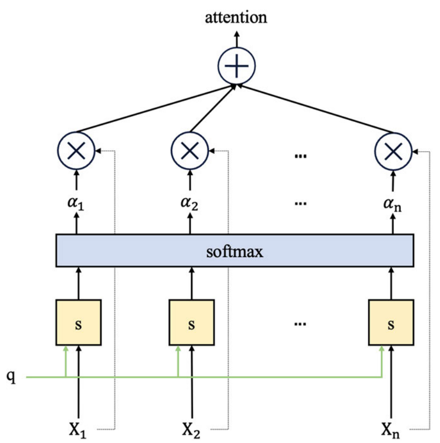

FIGURE 3. Soft attention mechanism structure.

Soft Attention Mechanism: For the i-th input information, given the query vector q and the input information vector X_i, the formula for selecting the probability α_i for the i-th input information is expressed as shown in Equation 2:

$$\alpha_i = p\left(z = i \mid X, q\right) = softmax\left(s\left(X_i, q\right)\right) = \frac{\exp\left(s\left(X_i, q\right)\right)}{\sum_{i=1}^{N} \exp\left(s\left(X_i, q\right)\right)} \quad (2)$$

The probability vector formed by α_i is called the attention distribution, and $s\left(X_i, q\right)$ represents the scoring function for attention. The scoring function for a linear model is defined as shown in Equation 3.

$$s\left(X_i, q\right) = X_i^T W q \quad (3)$$

Therefore, the attention distribution α_i indicates the relevance of the i-th information in the input information vector X_i to the given query q. By using Equation 4, the value of attention is obtained:

$$attention\left(X, q\right) = \sum_{i=1}^{N} \alpha_i X_i \quad (4)$$

D. ADAMK OPTIMIZER

Training GAN models poses the challenge of difficult convergence. Generative Adversarial Networks (GANs) require both the generator and the discriminator to reach a state of equilibrium during the adversarial process. The model may achieve a balanced state through multiple rounds of adversarial play. However, mutual cancellation between the generator and discriminator does not necessarily lead to the convergence points being simultaneously optimal. Therefore, the model exits when reaching an optimal state and initiates a new round of adversarial play. This adversarial process represents a non-convex problem, where non-convexity signifies the presence of multiple extrema within this time interval, indicating the existence of multiple stable equilibrium states. Hence, the AdamK adaptive optimization algorithm, based on gradient difference correction, is more suitable for the training mode of TK-GAN.

The Adam algorithm [37] (Adaptive Moment Estimation) is fundamentally an RMSprop algorithm with a momentum term. It dynamically adjusts the learning rates for each parameter using the first and second-order moment estimates of the gradients. The advantage of the Adam algorithm lies in the fact that, after bias correction, the learning rate for each iteration falls within a determined range, promoting parameter stability. It can be represented as follows:

$$\begin{cases} m_t = \beta_1 m_{t-1} + \left(1 - \beta_1\right) g_t \\ v_t = \beta_2 v_{t-1} + \left(1 - \beta_2\right) g_t^2 \\ \hat{m}_t = \dfrac{m_t}{1 - \beta_1^t}, \hat{v}_t = \dfrac{v_t}{1 - \beta_2^t} \\ \theta_{t+1} = \theta_t - \dfrac{\alpha}{\sqrt{\hat{v}_t + \epsilon}} \hat{m}_t \end{cases} \quad (5)$$

In the equation, m_t and v_t represent the first-order and second-order moment terms, respectively. $\beta_1 and \beta_2$ are momentum values, commonly set to 0.9 and 0.999, respectively. $\hat{m}_t \ and \ \hat{v}_t$ are the bias-corrected first-order and second-order moment terms. θ_t denotes the model parameters at iteration t, and ϵ is a smoothing value added to prevent division by zero. ϵ is typically set to a very small number, such as 1e-8, to ensure computational stability.

The momentum in the Adam algorithm is directly included in the first-moment (exponentially weighted) gradient estimate. Additionally, unlike RMSProp, Adam includes bias correction for both the first moment (momentum term) and the (non-central) second moment. While occasionally requiring adjustments to the learning rate from the suggested default values, Adam is generally considered robust when selecting hyperparameters and is applicable to most non-convex optimizations, large datasets, and high-dimensional spaces.

TABLE 3. Adam algorithm.

Adam algorithm: All vector calculations are corresponding element calculations
Input: Learning rate α_0 , First-order moment decay rate β_1 , Second-order moment decay rate β_2 , Stochastic objective function $f(\theta)$ The random initialization parameter vector θ_0 Initialize the parameter vector $m_0 = 0$， $v_0 = 0$， Initialize time step $t = 0$ While θ_t misconvergence do $t \leftarrow t + 1$ calculate the gradient of the stochastic objective function at time step t: $g_t \leftarrow \nabla_\theta f(\theta_t)$ Update the first-order moment estimate of the gradient： $m_t \leftarrow \beta_1 * m_{t-1} + (1-\beta_1) g_t$ Update the second-order moment estimate of the gradient： $v_t \leftarrow \beta_2 * r_{t-1} + (1-\beta_2) {g_t}^2$ If AMSGrad then： $v_t \leftarrow max(v_t, v_{t-1})$ Update parameter： $\hat{m}_t = \frac{m_t}{1-{\beta_1}^t}$, $\hat{v}_t = \frac{v_t}{1-{\beta_2}^t}$, $\theta_{t+1} \leftarrow \theta_t - \left(\frac{\alpha}{\sqrt{\hat{v}_t}+\varepsilon}\right)\hat{m}_t$

TABLE 3 shows the overall process of the Adam algorithm. Let $f(\theta)$ 为 be the differentiable objective function with respect to θ. This paper aims to minimize the expected value of this objective function with respect to the parameters θ, using $f(\theta_1)$, $f(\theta_2)$, $\cdots f(\theta_t)$ to represent the random objective functions corresponding to time steps 1,2, ..., t. This randomness comes from the random sampling of each mini-batch or inherent noise in the function. The gradient is denoted as $g_t = \nabla_\theta f(\theta_t)$, representing the partial derivative vector of the objective function f_t with respect to the parameters θ at time step t. The Adam algorithm performs exponential moving averages only on g_t and g_t^2. If AMSGrad is used, v_t must be constrained to be monotonically increasing, where $\frac{\alpha}{\sqrt{\hat{v}_t}+\varepsilon}$ is used as the adaptive update step size multiplied by $\hat{m}_t$ to update the parameters θ. If the gradients are larger over a period of time steps, the exponential moving average of g_t^2, denoted as $\hat{v}_t$, increases, leading to a smaller adaptive update step size. Conversely, if the exponential moving average of g_t^2 is smaller, $\hat{v}_t$ decreases, resulting in a larger adaptive update step size.

The proposed AdamK algorithm improves the Adam algorithm without significantly increasing time and space complexity. Table 4 illustrates the overall process of the AdamK algorithm. Unlike the Adam algorithm, AdamK uses $\Delta g_t = g_t - g_{t-1}$ to represent the gradient difference between time steps t and $t-1$. In contrast to the Adam algorithm, which performs exponential moving averages only on g_t and g_t^2, the AdamK algorithm performs exponential moving averages on g_t, Δg_t, and g_t^2, using the moving average of the latter to correct the former. If AMSGrad is used, v_t must be constrained to be monotonically increasing. Then, a bias correction is applied to obtain $\hat{z}_t$ and $\hat{v}_t$, where $\frac{\alpha}{\sqrt{\hat{v}_t}+\varepsilon}$ is used as the adaptive update step size multiplied by $\hat{z}_t$ to update the parameters θ. Additionally, this paper moves the constant ε from outside to inside the square root, as shown in Equation 6 and Equation 7.

$$\theta_{t+1} = \theta_t - \left(\frac{\alpha}{\sqrt{\hat{v}_t + \varepsilon}}\right)\hat{m}_t \tag{6}$$

$$\theta_{t+1} = \theta_t - \left(\frac{\alpha}{\sqrt{\hat{v}_t + \varepsilon}}\right)\hat{m}_t \tag{7}$$

As the iterations continue, when the parameter θ_t approaches the optimal value, the parameter gradient will become gradual, and the corrected second-order moment $\hat{v}_t$ tends to zero. At this point, the learning rates under the two different calculation methods can be approximately considered as $\frac{\alpha}{\varepsilon}$ and $\frac{\alpha}{\sqrt{\varepsilon}}$. Since the size of the small constant term is typically set to 10^{-8} or even smaller, then $\sqrt{\varepsilon} \gg \varepsilon$, and $\frac{\alpha}{\sqrt{\varepsilon}} \ll \frac{\alpha}{\varepsilon}$. In the later stages of iterations, reducing the learning rate helps improve the stability and convergence of optimization, with the latter having a more stable advantage. Based on the above analysis, a corresponding conclusion can be drawn: by incorporating the negligible term ε into the update of $\hat{v}_t$, it is possible to decrease the learning rate in the later stages of iterations, thereby enhancing the stability and convergence of the optimization algorithm.

TABLE 4. AdamK algorithm.

AdamK Algorithm: All vector calculations are element-wise
Input: Learning rate α_0， First moment exponential decay rate β_1， Second moment exponential decay rate β_2， Stochastic objective function $f(\theta)$ Randomly initialize the parameter vector θ_0 Initialize the parameter vector $m_0 = 0$， $v_0 = 0$， $r_0 = 0$ Example Initialize the time step $t = 0$ While θ_t misconvergence do IF $t = 1$， then $\Delta g_t = 0$， else $\Delta g_t = g_t - g_{t-1}$ $t \leftarrow t + 1$ Calculate the gradient of the stochastic objective function over the t time step： $g_t \leftarrow \nabla_\theta f(\theta_t)$ Update the first moment estimates of gradient and difference： $m_t \leftarrow \beta_1 * m_{t-1} + (1-\beta_1) g_t$， $r_t \leftarrow \beta_1 * r_{t-1} + (1-\beta_1)\Delta g_t$ Second moment estimation of updating gradient： $v_t \leftarrow \beta_2 * r_{t-1} + (1-\beta_2) {g_t}^2$ The gradient pair estimation is modified by the differential first moment estimation： $z_t \leftarrow m_t + \beta_{1*} r_t$ IF AMSGrad then： $v_t \leftarrow max(v_t, v_{t-1})$ Update parameter： $\hat{z}_t = \frac{z_t}{1-{\beta_1}^t}$, $\hat{v}_t = \frac{v_t}{1-{\beta_2}^t}$, $\theta_{t+1} \leftarrow \theta_t - \left(\frac{\alpha}{\sqrt{\hat{v}_t+\varepsilon}}\right)\hat{z}_t$

III. EXPERIMENTAL RESULTS AND ANALYSIS

A. EXPERIMENTAL DATASET AND ENVIRONMENT

This article focuses on the Guizhou Maotai (600519) stock and constructs a longitudinal comparative experimental

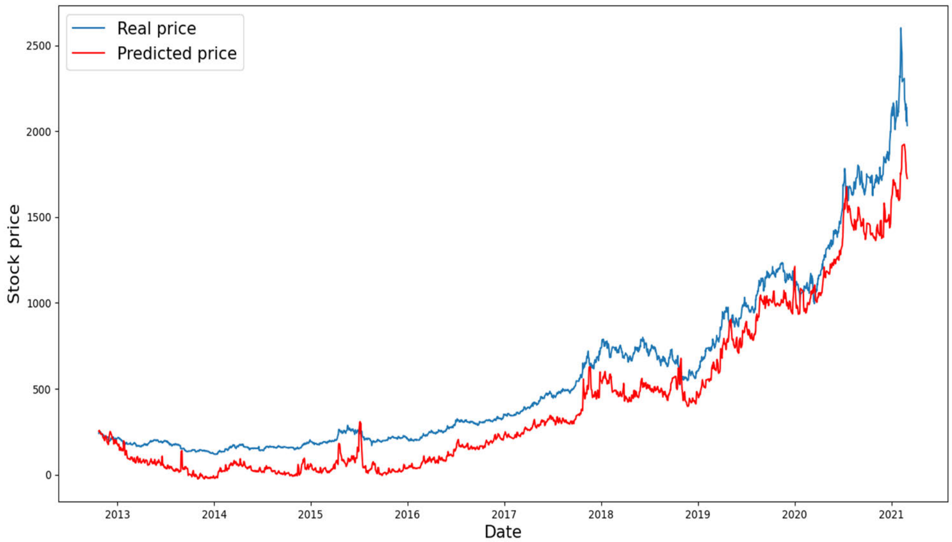

FIGURE 4. BiLSTM model test set prediction results.

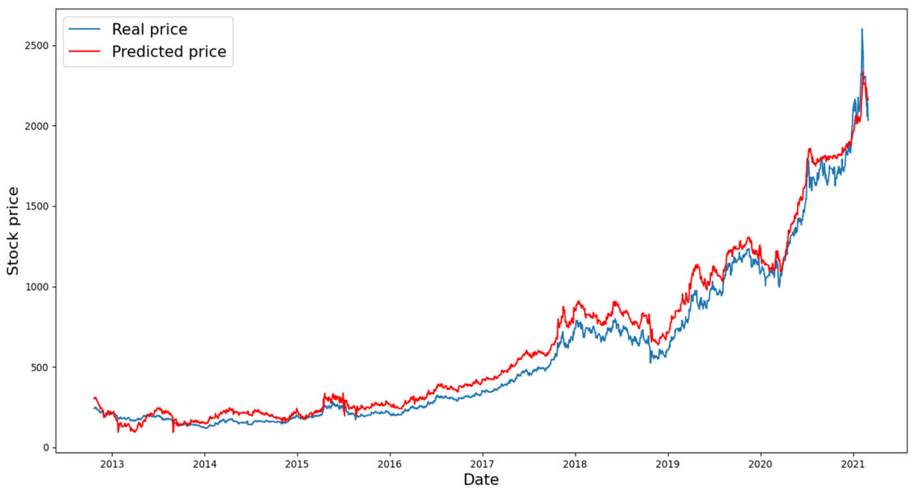

FIGURE 5. CNN model test set prediction results.

dataset. A sliding window of size 3 days is used, employing the numerical indicator data from the previous 3 days to predict the stock price on the 4th day. The sliding window moves forward by 1 day each time, predicting the stock price for the subsequent day. The iteration continues on the entire dataset until a complete set of data is obtained. Corresponding textual research report data will be acquired in the same manner as before and undergo data preprocessing.

The performance of stocks is often influenced by the direction of fund flow, with large-scale inflows and outflows significantly impacting stock price trends. Therefore, this article incorporates relevant fund flow data into the predictive model to enhance the model's sensitivity to short-term data fluctuations. Additionally, bank interbank lending rates (LIBOR), which well reflect changes in market liquidity, and the exchange rates of the Euro and US Dollar to Chinese Yuan, commonly used in international trade, are chosen as predictive factors. This inclusion considers the impact of market liquidity changes and international market trade on stock prices, further improving the accuracy of the predictive model.

The complete set of numerical indicator data includes aspects such as market trends (closing price, highest price, lowest price, opening price, previous closing, change amount, change rate, turnover rate, trading volume, trading amount, total market value, circulating market value, and amplitude), funds, market, exchange rates, and technical indicators (7-day moving average, 21-day moving average, MACD, midline, upper line, lower line, sliding average, momentum, fast Fourier transform, absolute value of fast Fourier transform, and angle of fast Fourier transform). The time range of these data is from October 23, 2012, to March 4, 2021, and they are

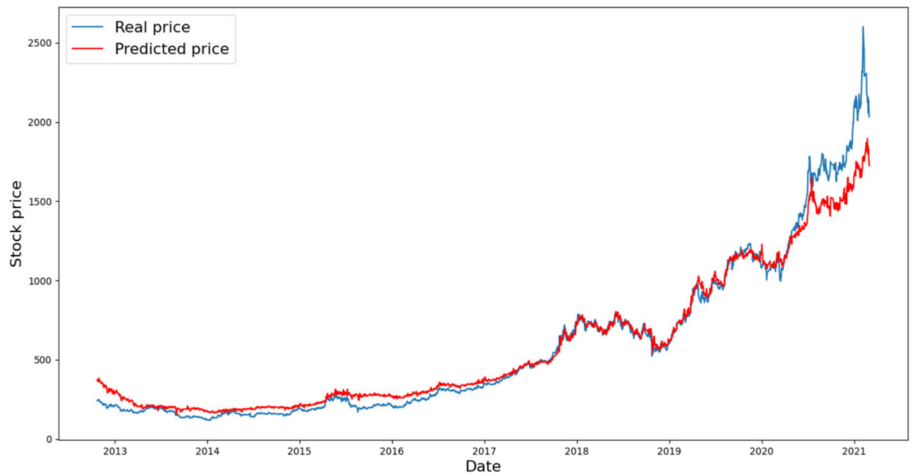

FIGURE 6. GRU model test set prediction results.

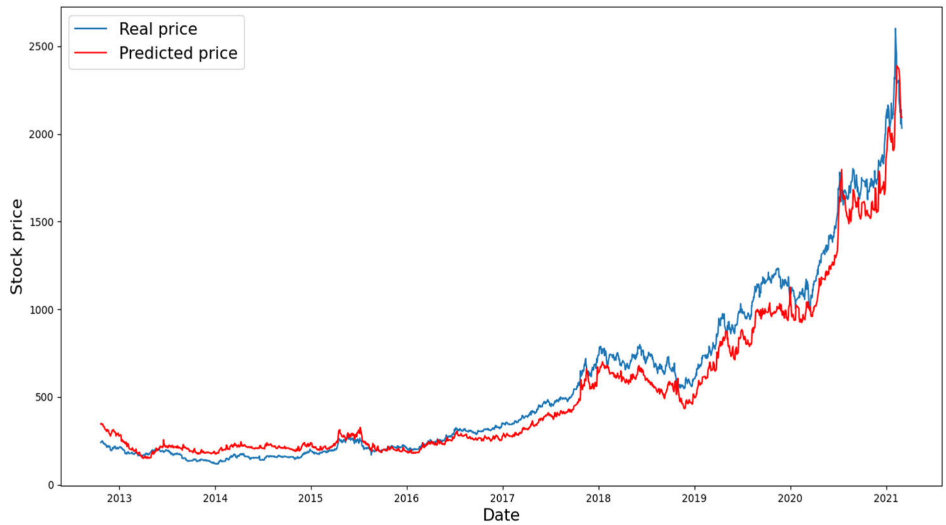

FIGURE 7. CNN-BiLSTM model test set prediction results.

divided into training, validation, and test sets in a 6:2:2 ratio. The final breakdown of the experimental dataset is shown in TABLE 5.

The experimental operating system utilized in this study is Windows 10, with an Intel(R) Core(TM) i9-10900K CPU @ 3.70GHz 3.70GHz processor and a GeForce RTX 3080 GPU, running on the Pytorch framework. The initial learning rate is set at 0.000002 and the model undergoes training for a total of 200 epochs.

B. MODEL EAVLUATION METRICS

To assess the predictive performance of the model, three regression evaluation metrics are employed: Mean Absolute Error (MAE), Mean Square Error (MSE), and Root Mean Square Error (RMSE). The formulas for calculating these three metrics are presented in Equations 8 to 10:

$$MAE = \frac{1}{m}\sum_{i=1}^{m}\left|\left(y_i - \hat{y}_i\right)\right| \tag{8}$$

TABLE 5. Experimental dataset division.

Type	Type Name	Dimension size	Time span
Numerical indicator data	X_train	（1218, 3,37）	2012/10/23 – 2017/10/27
	X_valid	（406, 3, 37）	2017/10/27 – 2019/07/02
	X_test	（406, 3, 37）	2019/07/02 – 2021/03/04
	y_train	（1218, 1）	2012/10/23 – 2017/10/27
	y_valid	（406, 1）	2017/10/27 – 2019/07/02
	y_test	（406, 1）	2019/07/02 – 2021/03/04
Seminar Text Data	text_train	（1218,3,1759）	2012/10/23 – 2017/10/27
	text_valid	（406, 3, 1759）	2017/10/27 – 2019/07/02
	text_test	（406, 3, 1759）	2019/07/02 – 2021/03/04

$$MSE = \frac{1}{m}\sum_{i=1}^{m}\left(y_i - \hat{y}_i\right)^2 \tag{9}$$

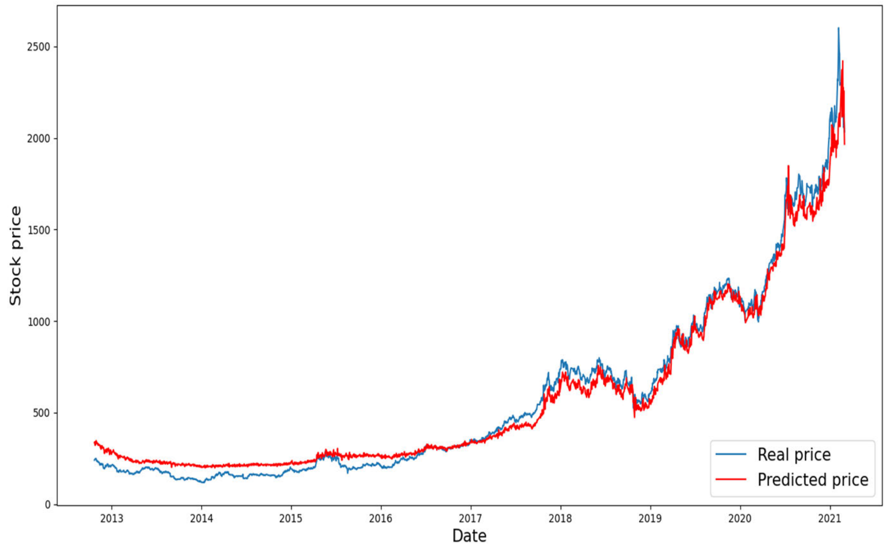

FIGURE 8. GAN model test set prediction results.

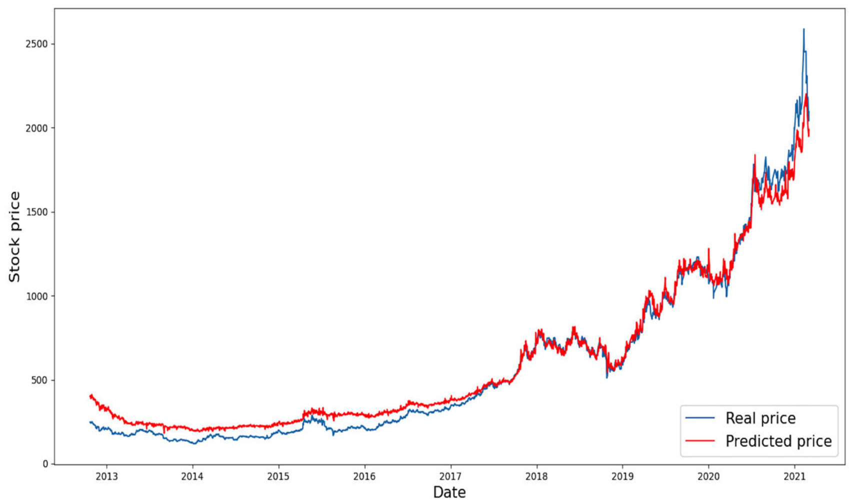

FIGURE 9. TEA model test set prediction results.

$$RMSE = \sqrt{\frac{1}{m}\sum_{i=1}^{m}\left(y_i - \hat{y}_i\right)^2} \tag{10}$$

In this context, m represents the number of samples, y_i denotes the actual (true) value, and $\hat{y}_i$ represents the predicted value. For MAE, MSE, and RMSE, smaller values indicate that the predicted values are closer to the actual values.

C. TIME WINDOW EXPLORATION

The TK-GAN model predicts the stock data value for the T + 1 day based on the stock data from the previous T days. However, the choice of the time window T can impact the model's predictive performance. To select an appropriate time window, controlled experiments are conducted on the TK-GAN model. In this study, time windows of t = 1, 2, 3, 4, and 5 days are individually selected, and the model's predictive performance is tested. The results of the experiments are as follows.

From TABLE 6, it is evident that when T = 1 and 5, the predicted data deviates more from the actual data. Choosing a time window of T = 3 days yields the best predictive performance, with MAE and MSE reaching 0.0195 and 0.0009, respectively.

D. COMPARISON WITH EXISTING ALGORITHMS

In order to validate the performance of TK-GAN model, this paper constructs a variety of common models and conducts training tests on Guizhou Maotai dataset to compare the experiments with TK-GAN model. The comparison

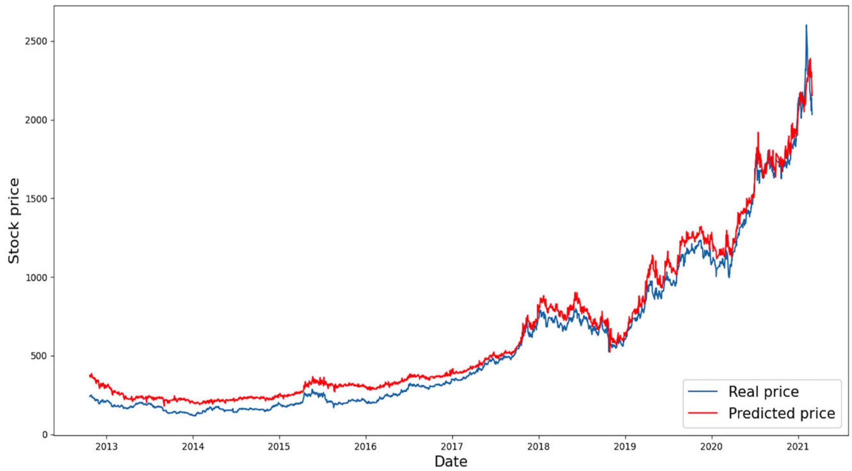

FIGURE 10. GAN- HPA model test set prediction results.

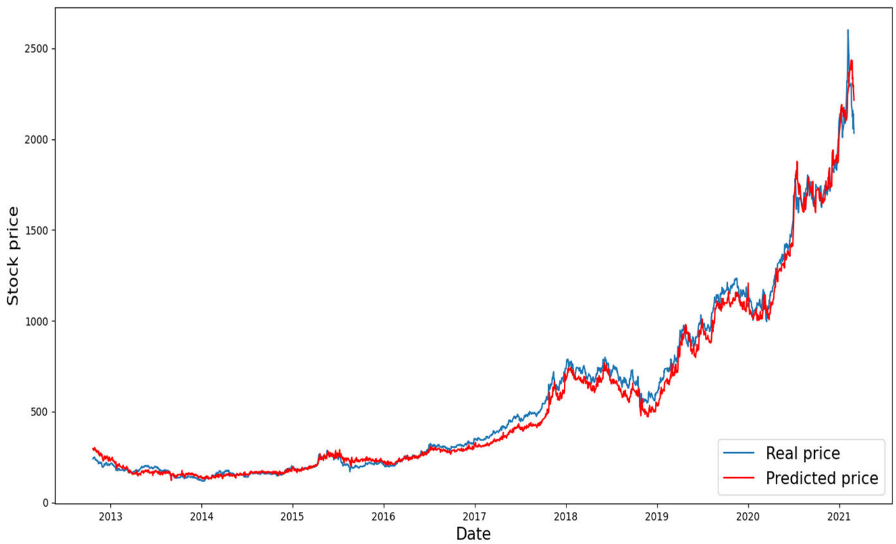

FIGURE 11. TK-GAN model test set prediction results.

TABLE 6. Comparative experiment results for time windows (Based on AdamK Optimizer).

Time window	MAE	MSE	RMSE
T=1	0.0478	0.0059	0.0768
T=2	0.0236	0.0014	0.0368
T=3	0.0195	0.0009	0.0301
T=4	0.0222	0.0010	0.0316
T=5	0.0529	0.0085	0.0923

models include: CNN, BiLSTM, CNN-BiLSTM, GRU, GAN, TEA and GAN-HPA models.In the GAN model, BiLSTM is selected for the generator, and CNN is selected for the discriminator.The experiments use the MAE, MSE, and RMSE as the evaluation indexes of the models, and the specific experimental values are shown in TABLE 7.

As shown in TABLE 7, the TK-GAN model demonstrates a unique advantage in stock closing price prediction.GAN has a significant advantage in prediction accuracy compared to CNN, BiLSTM and GRU networks. Compared with GAN, TEA and GAN-HPA, TK-GAN proposed in this paper has smaller evaluation indexes of MAE, MSE and RMSE, with only 0.02, 0.0009, and 0.0306. The experimental results show that the TK-GAN model is better in predicting the closing price of stock indices, closer to the real value and with smaller prediction error. The following figures show the closing price prediction results of Guizhou Moutai stock dataset, where FIGURE 4, 5, 6, 7, 8, 9, 10 and 11 are the closing price prediction results of the stock using BiLSTM, CNN, GRU,

TABLE 7. Experimental evaluation metrics for different prediction models (Based on Adam Optimizer).

Model	MAE	MSE	RMSE
CNN	0.0262	0.0017	0.0412
BiLSTM	0.0700	0.0093	0.0963
CNN-BiLSTM	0.0243	0.0015	0.0393
GRU	0.0303	0.0021	0.0456
GAN	0.0250	0.0013	0.0363
TEA[14]	0.0231	0.0012	0.0359
GAN-HPA[20]	0.0224	0.0012	0.0348
TK-GAN	0.0200	0.0009	0.0306

CNN-BiLSTM, GAN, TEA, GAN- HPA, and TK-GAN models, respectively.

E. OPTIMIZER COMPARATIVE EXPERIMENT

To further assess the impact of the AdamK optimizer on the performance of the TK-GAN model, different optimizers are applied to the TK-GAN model. In the optimizer comparative experiment, the TK-GAN model structure is used, and all hyperparameter values are standardized to control experimental variables. Additionally, to ensure experiment replicability and interpretability, a consistent random seed is employed. The experimental results for the TK-GAN model based on different optimizers are presented in TABLE 8.

TABLE 8. Experimental results for TK-GAN model with different optimizers.

Optimizer	MAE	MSE	RMSE
Adam	0.01999	0.00094	0.03063
AdamW	0.02729	0.00149	0.03857
SGD	0.27571	0.14535	0.38125
AdamK	**0.01949**	**0.00091**	**0.03014**

From TABLE 8, it is evident that the predictive experiment for the TK-GAN model based on the novel AdamK optimizer achieves optimal results. This indicates that AdamK is more conducive to enhancing the performance of the TK-GAN model compared to other optimizers.

IV. CONCLUSION

The proposed TK-GAN model, based on generative adversarial networks, incorporates sentiment factors, the Soft Attention mechanism, BERT pre-trained models, and the AdamK optimizer for stock price prediction. Experimental validation on the Guizhou Maotai stock dataset demonstrates that the TK-GAN model, coupled with the AdamK optimizer, exhibits smaller error values.

Although this article incorporates text data, the data types are still relatively thin. In future research, additional financial statement data and stock volatility charts could be incorporated as multiple data sources. By handling cross-modal data, better feature capturing can be achieved, leading to an improvement in the overall predictive performance.

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